

pen simultaneously. (How this is done will be explained later.) This is the norm for most forms of communication today.

Each call on a mobile phone network takes place on a different channel.

The Cell

Before the Cellular Network, scientists had two problems they needed to solve. First, how does one reach out to an entire city while still not requiring mobile transmitters to be ugly behemoths? And secondly, how can one ensure that all the people on the network can make calls from their mobile phones without exhausting the number of channels that can be supported in the system? Operators would only be authorised to use a set range of frequencies. A single antenna for an entire city would mean, say, 20 calls at a time if the authorised range could be divided into those 20 channels.



How frequency re-use would work for hexagonal, triangular and square cells. (No prizes for guessing the winner)

The answer to both these problems came in the form of the *cell*. A cell is a small area of service within a city, serviced by its own antenna. If one were to divide cities into little cells, the entire city could be easily covered. And each cell having its own antenna meant that the mobile phone transmitter need not to be that powerful - just powerful enough to reach the *base-station* that serviced the cell one was in.